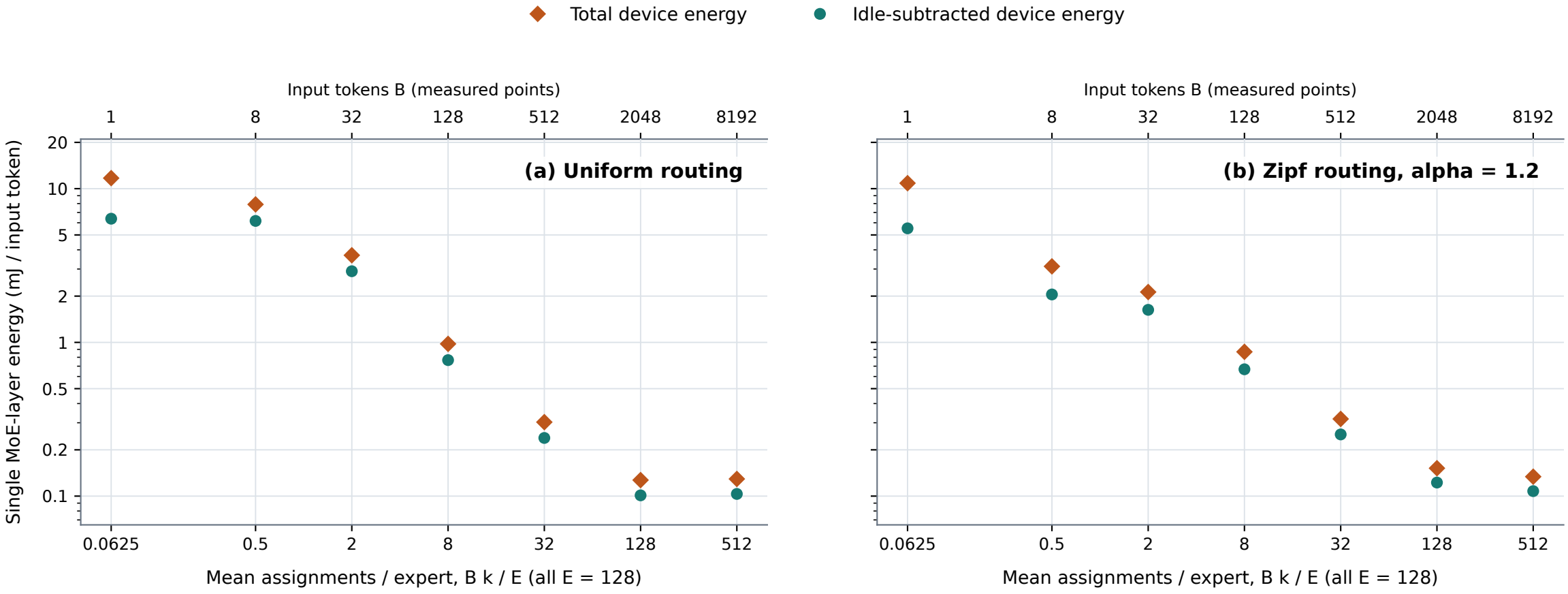
Qwen3 calibrated NVFP4: device energy per input token

NVIDIA checkpoint layer 0 | CUTLASS W4A4 | Same Qwen3 shape and CUDA graph replay | Energy protocol v2 only



NVFP4 energy v2 only; older BF16/FP8 energy v1 is excluded from comparison.

Primary measurement: NVML total-energy counter; per-case idle baseline, continuous-workload warmup, then a ~3 s window.

One energy window per point; no repeated-window uncertainty is available. This is one MoE layer, not whole-model J/token.

Points span measured B = 1-8192 (mean assignments 0.0625-512); no interpolation, quality claim, or serving gain. AWQ absent.